Data Analytics Project: IGN Game Reviews Analysis

Dataset Used: ign.csv

This document outlines a structured, end-to-end data analysis project based on game reviews from the IGN dataset. The project is divided into five scenarios, increasing in difficulty from basic data handling to advanced analysis and visualization, ensuring comprehensive skill coverage.

Project Goal: To analyze game review scores, platform popularity, and genre distribution over time using the `ign.csv` dataset.

The **primary libraries/modules** required for this project are:

- **Pandas:** For all data manipulation, filtering, aggregation, and feature engineering tasks.
- **NumPy:** Specifically for numerical calculations like yearly growth (`np.diff`).
- **Visualization Library (e.g., Matplotlib):** For generating all required plots (Line Graphs, Bar Charts, Pie Charts, Stacked Bar Charts).

🎯 Project Coverage

Category	Skills Covered
Data Preprocessing	Data Loading, Missing Value Handling (Mean/Mode), Type Conversion, Column Removal
Pandas Operations	Grouping, Filtering, Aggregation, Feature Engineering (score_category, editors_choice conversion)
NumPy Calculations	Array conversion, Yearly score growth calculation (np.diff)
Visualization	Line Chart, Bar Chart, Pie Chart, Histogram, Stacked Bar Chart, Saving Plots

Scenario 1: Basic Data Loading & Cleaning (🟢 Easy)

Description:
This scenario focuses on loading and preparing the IGN dataset for analysis using Pandas. The dataset is explored by viewing the first and last rows and checking its shape. Unnecessary columns are removed to clean the data. Missing values in important columns are identified and handled properly. Data types are also corrected to ensure accurate analysis.

Task	Description
1.	Load the `ign.csv` dataset using Pandas.
2.	Display the first 5 rows (`head()`), last 5 rows (`tail()`), and the shape of the dataset.
3.	Remove the unnecessary column: `"Unnamed: 0"`.
4.	Check for missing values in `score`, `genre`, and `platform`.
5.	Handle missing values: Fill the numeric column `score` with the mean, and the categorical column `genre` with the mode.
6.	Ensure correct data types: Convert `score` → float and `release_year`, `release_month`, `release_day` → integer.

Output of Scenario 1:

```
   Unnamed: 0 score_phrase  ... release_month release_day
0           0      Amazing  ...             9          12
1           1      Amazing  ...             9          12
2           2        Great  ...             9          12
3           3        Great  ...             9          11
4           4        Great  ...             9          11

[5 rows x 11 columns]
       Unnamed: 0 score_phrase  ... release_month release_day
18620       18620         Good  ...             6          29
18621       18621      Amazing  ...             6          29
18622       18622     Mediocre  ...             6          28
18623       18623  Masterpiece  ...             6          28
18624       18624  Masterpiece  ...             6          28

[5 rows x 11 columns]
(18625, 11)
```

Conclusion:

The dataset was successfully loaded and cleaned. Missing values in the genre column were handled properly. Unnecessary columns were removed, and data types were corrected. The dataset is now ready for further analysis. This step ensures accurate results in later scenarios.

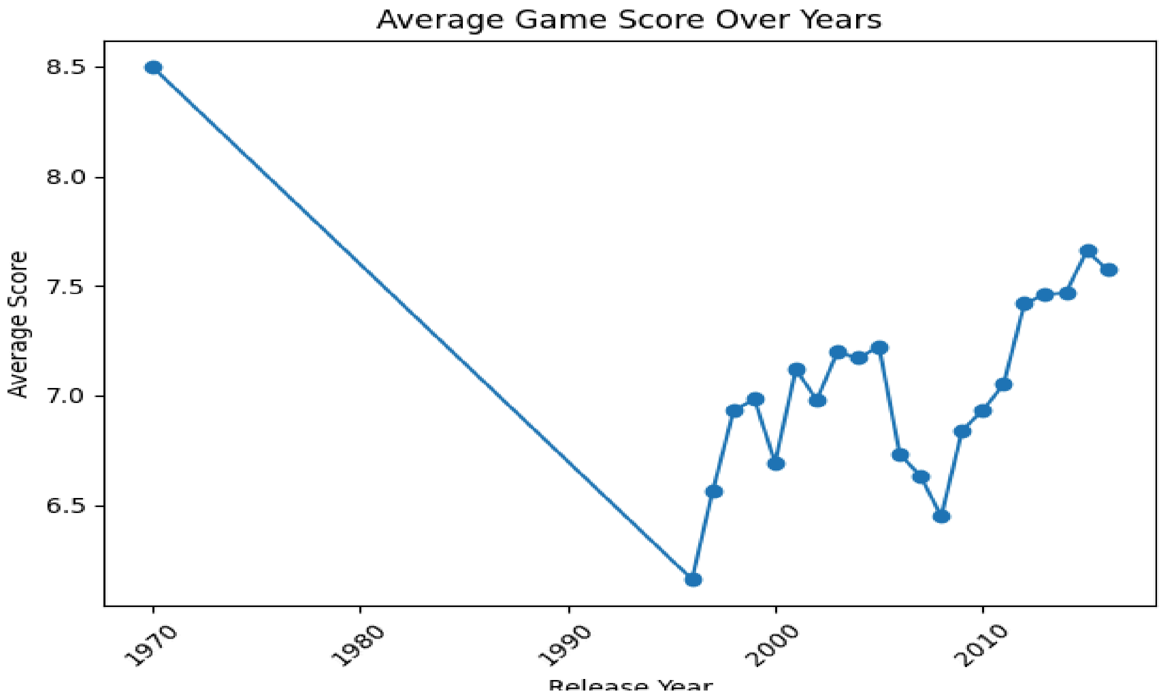
Scenario 2: Line Graph (Score Trend) + Save (🟢 Easy)

Description:

This scenario analyzes how game scores change over time. The data is grouped by release year, and the average score is calculated using Pandas. The results are visualized using a line graph created with Matplotlib. The graph helps in understanding trends in game scores across different years.

Task	Description
1.	Group data by `release_year`.
2.	Calculate the average score per year using Pandas.
3.	Convert the resulting average scores and years into NumPy arrays.
4.	Plot a line graph with `release_year` on the X-axis and average score on the Y-axis.
5.	Add the title: "Average Game Score Over Years" and proper axis labels.
6.	Save the graph: `plt.savefig("avg_score_trend.png")`.

Graph :
The graph is a line chart where the X-axis represents release years and the Y-axis represents average scores. Each point shows the average score for a year, and the points are connected by a line. The upward or downward movement of the line shows the trend of scores over time.



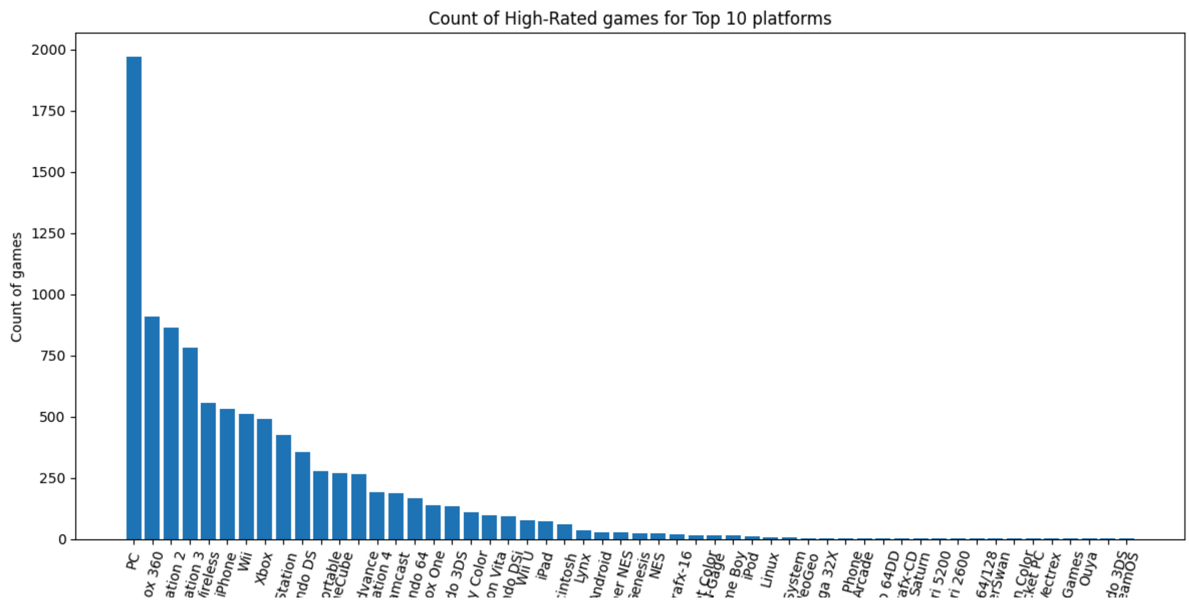
Conclusion:
The line graph shows how game scores have changed over the years. It helps in identifying whether scores are increasing or decreasing. The trend provides insight into the quality of games over time. Overall, it helps in understanding long-term patterns.

Scenario 3: Filtering + Bar Chart + Save (🟡 Medium)

Description:
This scenario focuses on comparing platforms based on high-rated games. The dataset is filtered to include games with scores greater than 7 using Pandas. The number of such games is counted for each platform. A bar chart is created using Matplotlib to visualize the top platforms.

Task	Description
1.	Filter the dataset to include only games where `score > 7`.
2.	Count the number of high-rated games per platform.
3.	Select the top 10 platforms using Pandas.
4.	Convert the platform counts into NumPy arrays.
5.	Plot a bar chart with `platform` on the X-axis and count of games on the Y-axis.
6.	Rotate x-axis labels for readability.
7.	Save the graph: `plt.savefig("top_platforms_bar.png")`.

Graph:
The graph is a bar chart where the X-axis represents platforms and the Y-axis represents the number of high-rated games. Each bar shows how many high-scoring games are available on that platform. Taller bars indicate better-performing platforms.



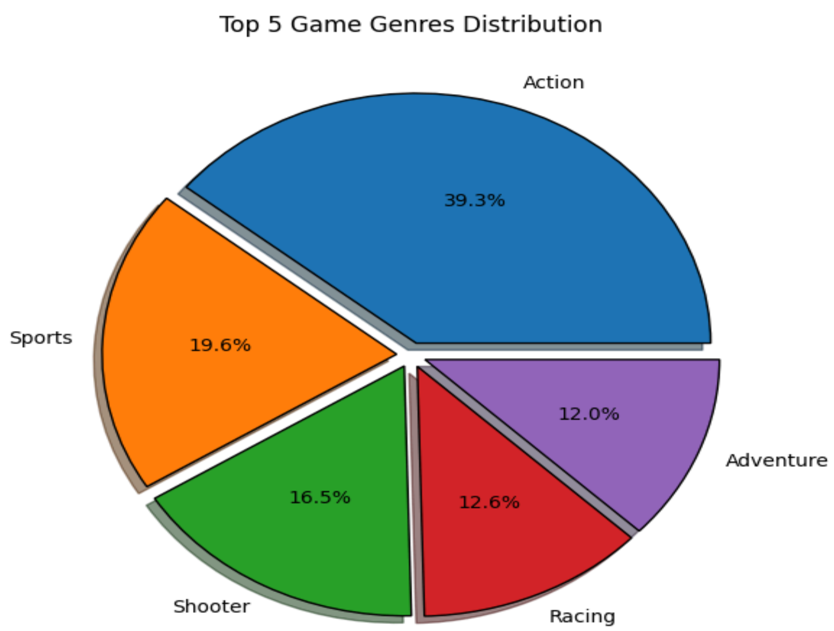
Conclusion:
The bar chart shows which platforms have the highest number of high-rated games. It helps in identifying the most successful platforms. Platforms with higher counts dominate in quality games. This provides useful insights into platform performance.

Scenario 4: Aggregation + Pie Chart + Save (🟡 Medium)

Description:
This scenario analyzes the distribution of games across different genres. The number of games in each genre is counted using Pandas. The top genres are selected and visualized using a pie chart with Matplotlib. This helps in understanding which genres are most common.

Task	Description
1.	Count the total number of games per `genre`.
2.	Select the top 5 genres using Pandas.
3.	Prepare labels (genres) and values (counts) for the pie chart.
4.	Plot a pie chart using `genre` as labels and `count` as values.
5.	Add percentage labels using `autopct` (e.g., '%.1f%%').
6.	Save the graph: `plt.savefig("genre_distribution.png")`.

Graph :
The graph is a pie chart where each slice represents a genre. The size of each slice shows the proportion of games in that genre. Percentage labels are added to make comparison easier. Larger slices indicate more popular genres.



Conclusion:
The pie chart shows the distribution of games across genres. It helps in identifying the most popular genres. Some genres contribute more than others. This gives a clear view of genre popularity.

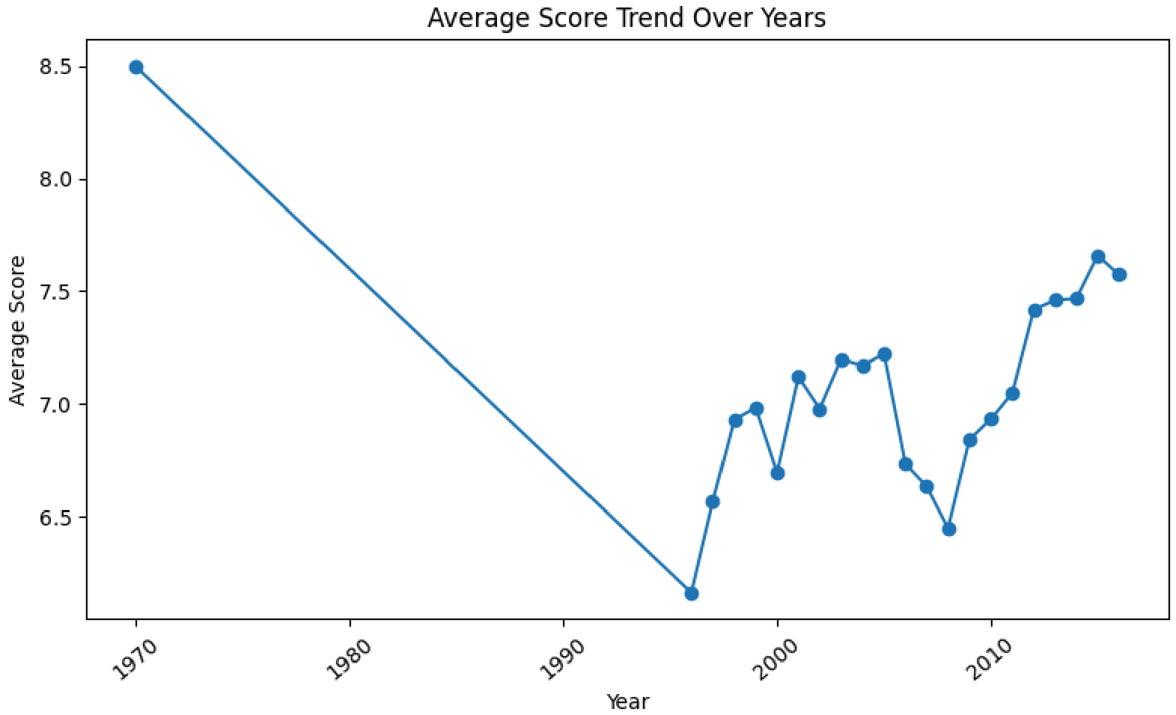
Scenario 5: Advanced Analysis + Multiple Graphs (🔴 Hard)

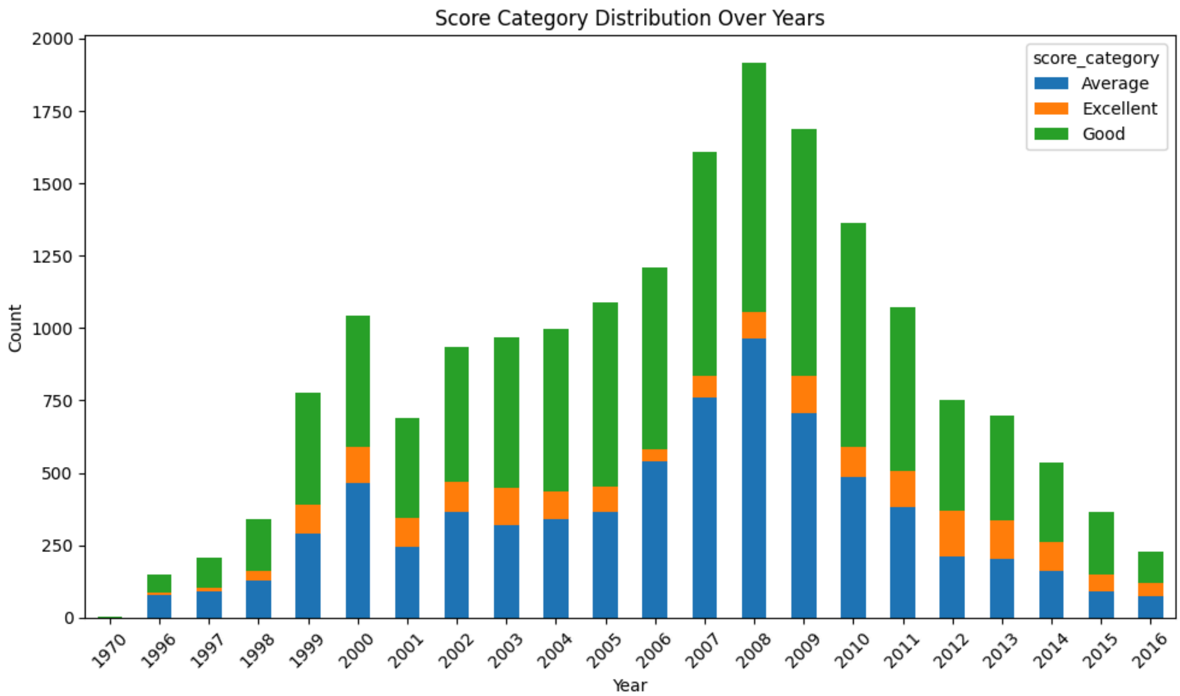
Description:
This scenario performs a detailed analysis of game review patterns. New features like score categories and numeric editor choice are created using Pandas. Yearly score trends and growth are calculated using NumPy. Multiple graphs such as line charts, stacked bar charts, and histograms are created using Matplotlib to visualize patterns and distributions.

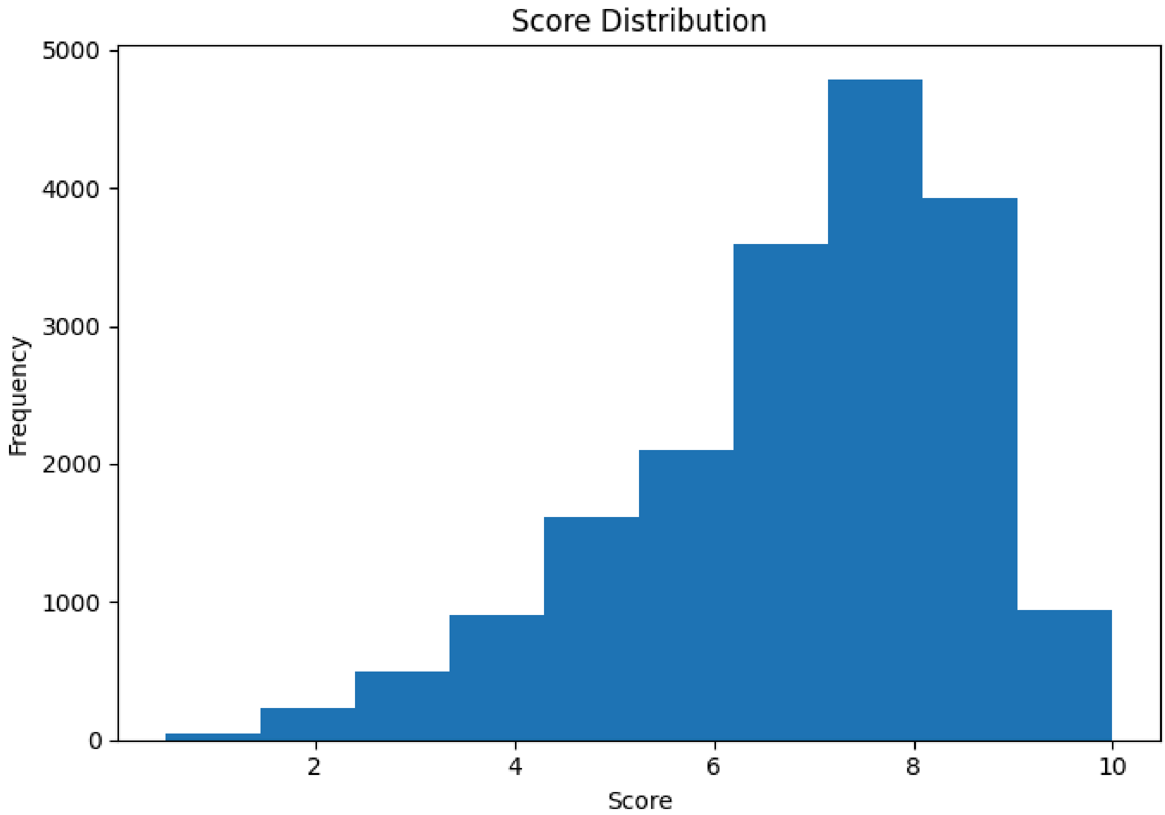
Task	Description
1.	Create a new column `score_category`: "Excellent" for score ≥ 9, "Good" for $7 \le$ score < 9, and "Average" for score < 7.
2.	Convert the `editors_choice` column: 'Y' $\rightarrow 1$, 'N' $\rightarrow 0$.
3.	Use NumPy's `np.diff()` to calculate the **yearly score growth** based on the average yearly scores from Scenario 2.
4.	Plot a **Line Graph** showing the trend of average score per `release_year`.
5.	Plot a **Stacked Bar Chart** showing the count of `score_category` per `release_year`.
6.	Plot a **Histogram** showing the distribution of `score`.
7.	Save the graphs: `plt.savefig("score_trend.png")`, `plt.savefig("score_category_stacked.png")`, `plt.savefig("score_distribution.png")`.
8.	Identify: Which years had the highest scores, whether high scores increased over time, and if `editors_choice` correlates with high scores.

Graph :

The **line graph** shows the trend of average scores over the years. The **stacked bar** chart shows how different score categories are distributed each year. The **histogram shows** how scores are distributed across different ranges. Together, these graphs provide a complete understanding of the data.







Conclusion:
The analysis shows how game scores and categories have changed over time. It helps in identifying years with high scores and overall trends. The results also show that higher scores are often associated with editor's choice. Overall, this scenario provides deep insights into game review patterns.

Final Conclusion:
This project analyzed game review data step by step. The dataset was cleaned and prepared for analysis. Different graphs like line chart, bar chart, and pie chart were used to understand trends and comparisons. The analysis showed patterns in game scores, platforms, and genres over time. Overall, the project provides a clear understanding of game review trends.

What you have learned
I learned how to clean and analyze data using Pandas. I used NumPy and Matplotlib for calculations and graphs. This project helped me understand data visualization and trends.